



From the archives: Calcium signaling and PIN polarity, functional divergence after genome duplication, and a 5' inhibitor of gene expression

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November 2021: Calcium signaling linked to PIN polarity

PIN-FORMED (PIN)-mediated polar auxin transport (PAT) forms auxin gradients that regulate various aspects of plant development. PAT is influenced by both internal and external signals, including calcium (Dela Fuente and Leopold, 1973), but the mechanisms underlying this process are largely unknown. Lee et al. (2021) showed that calcium-dependent protein kinase 29 (CPK29) phosphorylates PINs to affect auxin trafficking and polarity. Genetic evidence suggested that CPK29 is required to promote the exocytic trafficking of PINs for recycling. The authors proposed that CPK29 directly interprets internal and external Ca^{2+} signals to control plant development. Notably, CPK29 phosphorylates PINs at sites not targeted by other PIN-targeting kinases, including recently reported Shaggy/glycogen synthase kinase 3 (GSK3)-like kinases (AtSKs; Lu et al., 2022). Therefore, multiple parallel mechanisms maintain the PIN polarities and activities.

November 2017: Coordinated functional divergence after genome duplication

Whole genome duplication (WGD) is often detrimental to organismal fitness and an evolutionary “dead end.” But on some notable occasions, WGD has provided adaptive advantages, both short- and long-term, as evidenced by the existence of successful polyploid organisms around us (Van de Peer et al., 2017). WGD can increase plant fitness in a changing environment by providing extra “genes” that serve as substrates for the evolution of novel functions. However,

coordinated evolution of duplicated gene modules is rarely observed; so how do new pathways evolve?

To identify cases of polygenic evolution in plants, De Smet et al. (2017) analyzed the gene expression compendium of *Arabidopsis thaliana* that covers various tissues and conditions and identified 92 homoeologous genes (duplicated gene pairs) partitioned into two gene expression modules showing tissue-specific expression—one being expressed mostly in aerial tissues and the other predominantly in tip-growth tissues (root and pollen). Genes in the leaf module shared functions related to defense, while functions related to cell polarity and the maintenance of cell wall integrity were enriched in the root/pollen modules (Figure). Thus, the divergence of the gene modules in cell type-specific expression likely reflects functional divergence. The authors proposed that such network sub- or neofunctionalization driven by WGDs provided adaptations to environmental stresses.

November 1997: A 5' UTR inhibitor of gene expression

The 5' untranslated regions (UTRs) of mRNAs form various secondary or tertiary structures to control gene expression mainly via post-transcriptional regulation. In a few cases, 5' UTRs regulate transcription. Twenty-five years ago this month, Curie and McCormick reported such a rare function of a 5' UTR in plants. The authors found that the 5' UTR of LAT59 (late anther tomato 59), a pollen-specific gene, functions as a transcriptional suppressor without affecting translation. The DNA element within the 5' UTR contained a putative stem-loop, a secondary RNA structure known to